

Problem D: Cryptarithmic

Problem ID: cryptarithmic

Time Limit: 2 seconds

Memory Limit: 256 MB

A simple substitution cipher replaces each letter with another letter consistently. Given encrypted text and a dictionary, find the decryption mapping.

Input

The input consists of the following:

- The first line contains an integer n ($1 \leq n \leq 1000$), the number of words in the dictionary.
- The next n lines each contain a single word, consisting of lowercase English letters.
- Following the dictionary, the rest of the input consists of the encrypted text to be decrypted. The text may be spread across multiple lines and ends at the end of the file (EOF).

Output

Output the decrypted text. The case and punctuation of the original text should be preserved. If no unique valid decryption can be found, or if no solution exists, each letter of the encrypted text should be replaced with an asterisk (*).

Sample

Sample Input 1

```
6
and
dick
jane
puff
spot
yertle
bjvg xsb hxsx xsb qymm xsb rqat xsb pnetfn
xxxx yyy zzzz www yyyy aaa bbbb ccc dddddd
```

Sample Output 1

```
dick and jane and puff and spot and yertle
**** *  **** *  **** *  **** *  **** *  **** *
```